

Report 2

C Programming

About Report 2

- Please submit a summary of your Lecture 4,5,6,7, 8 and Problems in the following format:
 - Format: PDF
 - **Contents:**
 - Source code (Paste into a document with text, or paste with an image)
 - Execution result (screenshot of image)
 - Program description.
- **Submission Deadline: 30 June, 23:59**

***The use of generative AI is prohibited.**

Lecture 4 - 8, Summary about the following exercises:

- Lecture 4, Exercise 4: Tower of Hanoi
- Lecture 5, Exercise 7: Sorting(Bubble Sort)
- Lecture 6, Exercise 3: Structure
- Lecture 7, Exercise 3: Search
- Lecture 8, Exercise 3: Queued programs

In Report 2, you will need to summarize the exercises shown here.

Report 2

Problem 1: Reverse Polish Notation Calculator

- Create a Reverse Polish Notation Calculator using stacks.
- Reverse Polish notation is described as follows (in bold):

$1+2 \Rightarrow \mathbf{1\ 2\ +}$

$(2+3)*2 \Rightarrow \mathbf{2\ 3\ +\ 2\ *}$

*** Spaces between numbers and operators**

- Allow arbitrary expressions to be entered from the keyboard

Reverse Polish Calculator

How it works

input	stack	action
2	2	Push 2
2 3	2 3	Push 3
2 3 +	5	Pop 2 and 3 to push $2+3=5$
2 3 + 2	5 2	Push 2
2 3 + 2 *	10	Pop 5 and 2 to push $5*2=10$

```
$ ./a.exe _  
> 1 2 +  
3  
> 2 3 + 2 *  
10  
/
```

Report 2: Problem 2

Write a C program to print all Tribonacci numbers less than 10,000 using recursion.

Description:

- The Tribonacci sequence is a generalization of the Fibonacci sequence.
- It is defined as follows:
 - $T(0) = 0$
 - $T(1) = 1$
 - $T(2) = 1$
- For $n \geq 3$:
 - $T(n) = T(n-1) + T(n-2) + T(n-3)$

0 1 1 2 4 7 13 24 44 81 149 274 504 927 1705 3136 5768

Task:

- Write a C program that:
 - Implements a recursive function `trib(n)` that returns the n -th Tribonacci number.
 - Uses this function to print all Tribonacci numbers less than 10,000.
 - Stops printing when the next Tribonacci number would be 10,000 or greater.

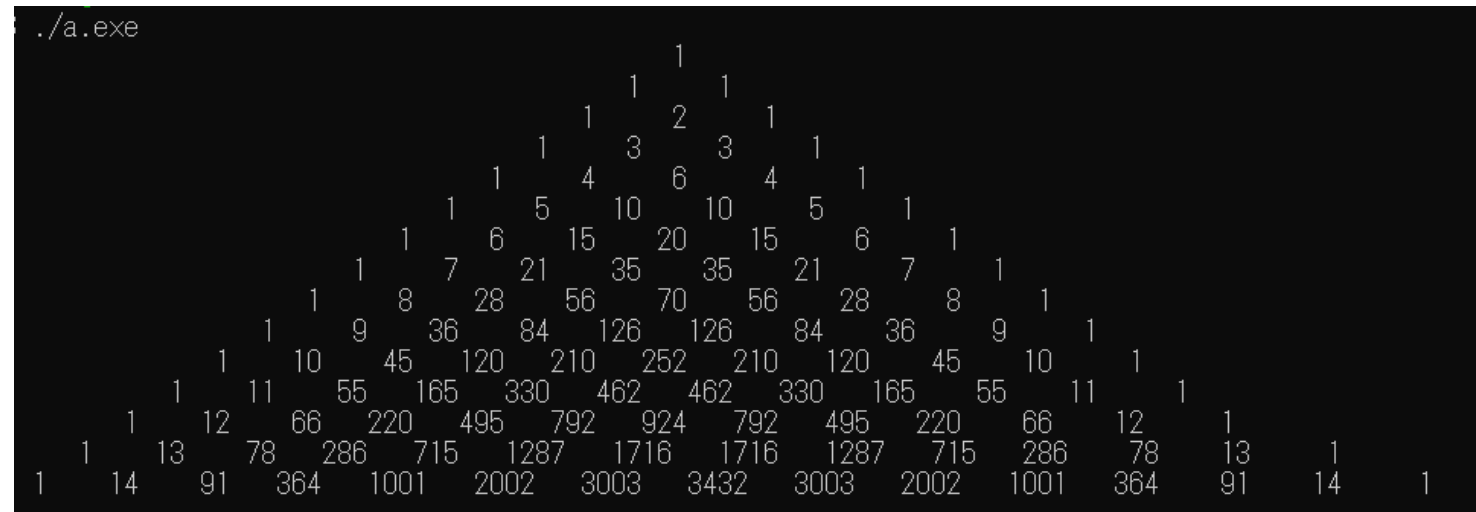
Report 2

Problem 3

Pascal's triangle is a triangular arrangement of the coefficients in the binomial theorem (binomial coefficients).

- Create a program to display 15 lines of this(Refer to the following image.)

1			
1	1		
1	2	1	



Examples of how-to summaries

The basic format for Report 2 is the same as Report 1.

Your Student number
Your Name
Lecture 1
Exercise1
Source Code
Execution result
Program description
Ex. Programs that display Hello.
Exercise 2
. . .
Lecture 2
Exercise1
. . .